

max heap sort :

```
def heapify(arr, n, i):
    largest = i
    left = 2 * i + 1
    right = 2 * i + 2

    # Check left child
    if left < n and arr[left] > arr[largest]:
        largest = left

    # Check right child
    if right < n and arr[right] > arr[largest]:
        largest = right

    # If largest is not root
    if largest != i:
        arr[i], arr[largest] = arr[largest], arr[i]
        heapify(arr, n, largest)

def heap_sort(arr):
    n = len(arr)

    # Build Max Heap
    for i in range(n // 2 - 1, -1, -1):
        heapify(arr, n, i)

    # Sort the array
    for i in range(n - 1, 0, -1):
        # Swap maximum element with last element
        arr[0], arr[i] = arr[i], arr[0]

        # Reduce heap size and heapify
        heapify(arr, i, 0)

# Input from user
n = int(input("Enter number of elements: "))
arr = list(map(int, input("Enter elements: ").split()))

heap_sort(arr)

print("Sorted array:", arr)
```

```
Enter number of elements: 5
Enter elements: 23 32 7 65 18
Sorted array: [7, 18, 23, 32, 65]
```

